

Coil Current and Vacuum Magnetic Flux Calculation for Axisymmetric Equilibria

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Abstract.

In fixed-boundary axisymmetric equilibrium calculations the plasma shape is assigned from input. In several circumstances, the plasma shape may not be known a priori, or one may desire to also compute the magnetic field in the volume surrounding the plasma through the calculation of a free-boundary equilibrium. This requires either the coil currents or the magnetic poloidal flux on a curve in the vacuum region to be assigned as input for the free-boundary equilibrium calculation. The FREE-FIX code presented in this article is a general tool for calculating coil currents being given a fixed-boundary calculation. A new formulation is presented, which considerably reduces the computational cost of the calculation. FREE-FIX performs well for different geometries and experiments.

Submitted to: *Plasma Phys. Control. Fusion*

1. Introduction and Problem Definition

Equilibrium calculation is a standard task in tokamak analysis. Under the assumption of axisymmetry, the equilibrium problem can be reduced to a two-dimensional elliptic PDE for the magnetic poloidal flux, which can assume different forms depending on the details of the equilibrium model. Small hyperbolic regions may be present in the plasma in special circumstances, [1, 2] which will not be discussed in this work. If the shape of the plasma edge and the value of the magnetic poloidal flux ψ (the main unknown of the problem) at the plasma edge are assigned from input, the equilibrium problem needs to be solved only in the plasma region. This type of calculation is known as “fixed-boundary” equilibrium calculation.

Given a “fixed-boundary” equilibrium, one may be interested in determining a posteriori the magnetic field in the vacuum region surrounding the plasma. In principle, if one obtains the coil currents corresponding to the desired plasma shape, one may be able to directly calculate ψ in the vacuum [see Eq. (3) in section 2]. However, as shown in detail in the remainder of this work, coil currents determined a posteriori for an arbitrary plasma shape will in general not reproduce the desired shape exactly. Thus, to have an accurate solution of the equilibrium problem in both the vacuum and plasma regions, one may have to recalculate the equilibrium in both regions and determine the plasma shape self-consistently as part of the solution of the equilibrium problem. This type of calculation is known as “free-boundary” equilibrium calculation. The problem addressed in this note and solved by the FREE-FIX code is how to determine the coil currents corresponding to a fixed-boundary

equilibrium being given (1) the fixed-boundary equilibrium and (2) the positions of the coils surrounding the plasma. If desired, the coil currents can then be used to generate the input for a free-boundary equilibrium calculation that will match as closely as possible the desired plasma shape. A free-boundary solver input can in general be reduced to two options: first, the value of the coil currents for each coil surrounding the plasma, or second, the value of the magnetic poloidal flux on a closed curve in the vacuum region. Given the coil currents or ψ on a closed curve outside the plasma (a standard post-processing output) from FREE-FIX, users can use their free-boundary equilibrium codes to calculate the free-boundary equilibrium. If a different output format is needed by a user, the necessary code development will be undertaken when required.

In principle, the vacuum field could be obtained using a different strategy, namely using the virtual casing principle [3, 4]. However, a virtual casing calculation requires the accurate knowledge of the poloidal magnetic field on the plasma boundary. Due to the Cartesian grid implementation of FLOW, [5, 6] (the code used for equilibrium calculations in this work) and other equilibrium codes, no grid points are situated on the plasma boundary and edge fields can be computed only with a lower accuracy than the fields inside the plasma, making the virtual casing approach impractical.

The scientific problems of tokamak free- and fixed-boundary equilibrium have been extensively studied in the past. In particular, Ref. [7] discusses issues related to the determination of coil currents for a given plasma shape and methods similar to the “standard” method discussed in section 2. One important point made in Ref. [7] is that one cannot calculate the vacuum field and flux extrapolating from the plasma edge. This creates the need for the calculation of the external coil currents described in this work.

In axisymmetric geometry and using a cylindrical (R, φ, Z) system of coordinates, the magnetic field is written as $\underline{B} = \nabla\psi \times \nabla\varphi + B_\varphi \hat{e}_\varphi$, where the toroidal field B_φ is in the direction of symmetry and the poloidal field $\underline{B}_p$ is the remaining part of the magnetic field. The poloidal magnetic flux ψ is the main unknown of the equilibrium problem. It is easy to show [8, 9] that

$$\Delta^*\psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -RJ_\varphi. \quad (1)$$

Equilibrium models can be cast in the form $\Delta^*\psi = \mathcal{RHS}$, where $\mathcal{RHS} = 0$ in the vacuum region surrounding the plasma (where no electrical current flows) and to some model-dependent expression in the plasma. If the goal of the user is a free-boundary equilibrium calculation, it is assumed that the input necessary to calculate the right hand side $\mathcal{RHS}$ inside the plasma will be provided by the user. FREE-FIX only provides the currents in the coils around the plasma and the value of ψ on a closed curve surrounding the plasma calculated from the coil currents and the fixed-boundary plasma current density distribution using Eq. (3) in Section 2. The methods implemented for this calculation are described next.

2. Solution Method, Old and New

The calculation of coil currents and vacuum ψ on the computational boundary is based on the Green’s function method. The problem has in fact a well-known solution, which

is discussed in detail e.g. in Ref. [10][p111]. The Green's function for the Δ^* operator is found in the literature [7, 10]:

$$G(R, Z; R', Z') = \frac{1}{2\pi} \frac{\sqrt{RR'}}{k} [(2 - k^2) K(k) - 2E(k)], \quad (2)$$

where $E(k)$ and $K(k)$ are complete elliptic integrals [11] and $k^2 \equiv 4RR'/[(R + R')^2 + (Z - Z')^2]$. Then, if the position of the coils, the coil currents I_i and the plasma shape and current density are known, ψ can be expressed at any point in the vacuum as

$$\psi(R', Z') = \int_P G(R, Z; R', Z') J_\varphi(R, Z) dR dZ + \sum_{i=1}^{N_c} G(R_i^c, Z_i^c; R', Z') I_i, \quad (3)$$

where the integral is taken over the plasma region, the sub- or super-script “c” indicates the N_c coils, which have been represented as wires of infinitesimal cross section for simplicity.

In order to calculate the coil currents, Eq. (3) is applied to points on the (prescribed) plasma boundary, under the reasonable assumption that no singular currents are present on the plasma boundary. Since the plasma boundary is defined by a curve Γ over which $\psi = \text{const.}$ (assumed to be $\psi = 0$ for simplicity), Eq. (3) evaluated in any (R', Z') point on Γ becomes a linear equation in the N_c unknowns corresponding to the coil currents. Even though the problem could in principle be solved directly as a linear system of equation evaluating Eq. (3) in N_c distinct points on Γ (the “direct method”), a minimization approach is needed to find a reasonable estimate of the coil currents. [7] Equation (3) is evaluated in a number of points on the plasma boundary $N_{fit} \gg N_c$ and the N_c currents that minimize the difference between the calculated and prescribed values of ψ on the N_{fit} points are found. Both a least square fit (LSF) and a singular value decomposition (SVD) method are implemented in FREE-FIX, giving undistinguishable results. In the cases tested in this work, which have a number of coils $N_c \sim 15$, a few tens to ~ 100 points were needed for satisfactory results (as opposed to the $N_c \sim 15$ that are used by the direct method). Both elliptic integral calculations and minimizations are implemented using Numerical Recipes [12] routines.

A different formulation, still relying on error minimization, but expressed in terms of equilibrium perturbation, can be found in Ref. [13].

The standard method described above imposes (in an LSF sense) the calculated $\psi = 0$ curve Γ_c to “meet” the desired curve Γ in a finite number of points, in the sense that the two curves go through the same points, but are not imposed to be tangent at the points where they meet. A stronger condition is to require Γ_c to be *tangent* to Γ . This was implemented in FREE-FIX, resulting in a second condition, assigned by calculating the unit vector $\hat{\tau}$ tangent to Γ and imposing $\nabla\psi \cdot \hat{\tau} = 0$. It is observed that

$$\nabla'\psi(R', Z') = \int_P \nabla'G(R, Z; R', Z') J_\varphi(R, Z) dR dZ + \sum_{i=1}^{N_c} \nabla'G(R_i^c, Z_i^c; R', Z') I_i \quad (4)$$

and that the necessary derivatives of the Green's function are given by

$$\begin{aligned} \frac{\partial G}{\partial R'} &= \sqrt{R} \left(\frac{\partial k}{\partial R'} R' (4K + k(2(E - 2K) + k(-kK + K + E)) - 4E) \right. \\ &\quad \left. - 2(k - 1)k (k^2 K - 2K + 2E) \right) / (4\pi(k - 1)k^2), \end{aligned} \quad (5)$$

$$\frac{\partial G}{\partial Z'} = \sqrt{R} \left(E((k(k+2) - 4) \frac{\partial k}{\partial R'} R' - 4(k-1)k) - (k-1) \right. \\ \left. ((k^2 + 4) \frac{\partial k}{\partial R'} R' + 2k(k^2 - 2)) K \right) / (4\pi(k-1)k^2), \quad (6)$$

where the argument k has been omitted from the elliptic integrals E and K for brevity. Remarkably, no derivatives of the equilibrium quantities are needed and the only elliptic integrals that need to be calculated are the same elliptic integrals that are needed in the calculation of the Green's function used in Eq. (3). This is especially useful because most of the computational cost in the calculation of the unknown coil currents I_c is in the evaluation of the elliptic integrals E and K . Thus, by imposing the two conditions $\psi = 0$ and $\nabla\psi \cdot \hat{\tau} = 0$ in each of N_{fit} points on Γ one needs about half the computational cost needed to impose $\psi = 0$ in $2N_{fit}$ points.

The “tangent” condition may appear to be only applicable to smooth boundaries, thus excluding boundaries with X-points. In truth, if the fixed-boundary equilibrium contains an X-point, one can assign two distinct directions for the tangent when coming arbitrarily close to the X-point from the two sides. This was implemented in FREE-FIX, but could not be tested thoroughly since FLOW does not explicitly allow for X-points on the boundary. Preliminary tests with an X-point added ad hoc at the lower edge of the fixed-boundary plasma did not show any convergence issue in the calculation of the coil currents.

The positioning of the fit points on the boundary influences the minimum number of points needed to obtain a good approximation of the desired shape. Two different strategies were found to produce similarly good results: First, the “equal distance” approach, which requires that the distance between two consecutive points be the same for any couple of consecutive points and Second, the “equal arc length” approach, which requires consecutive points to be equally spaced in the length of the curve surrounding the plasma (meaning that the distance between any two consecutive points measured along the curve surrounding the plasma is the same). A summary of the different methods and options used in this paper is contained in Ref. [14].

3. Sample Application

FREE-FIX was used to calculate the coil currents for equilibria of two of the major US tokamaks, NSTX [15] and DIII-D [16,17]. For each of the equilibria shown in this section, a fixed-boundary equilibrium with an assigned boundary shape was computed first. The calculated [using Eq. (3)] equilibrium ψ was then used as an input for FREE-FIX to calculate coil currents and thus ψ on a closed “box” around the plasma, needed as boundary condition by the equilibrium code FLOW. A new free-boundary equilibrium was computed and compared to the desired (fixed-boundary) one. For each experiment a set of coils approximating the real coil geometry was used. The input used for all the calculations presented in this section is available from FREE-FIX' website. [14]

The main findings of a sample “spherical-tokamak” NSTX equilibrium calculation are collected in Fig. 1, which shows the plasma shape determined by four free-boundary equilibria (in color) and compares them with the desired plasma shape (black). Figure 1(a) shows that the “direct method” does not produce satisfactory results. Reasonable accuracy is obtained from: an LSF FREE-FIX calculation using 100 points uniformly distributed in θ (the poloidal angle), Fig. 1(b); an SVD calculation using only seven points (for nine coil currents plus the central solenoid), in which both the $\psi = 0$

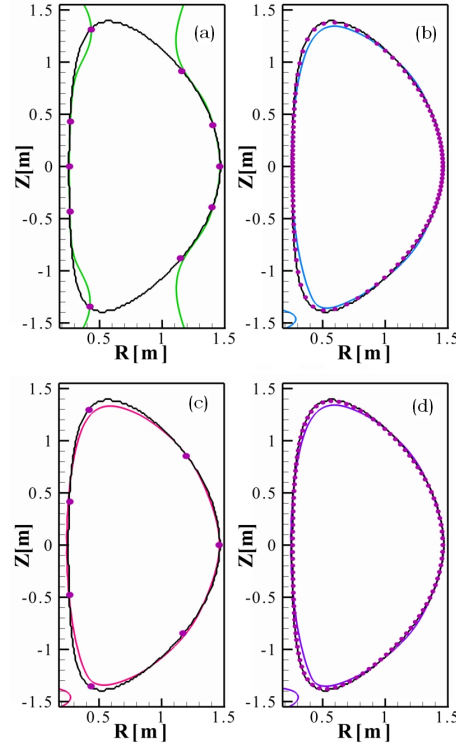


Figure 1. Desired plasma shape (black) and plasma shapes calculated with FREE-FIX input for an NSTX equilibrium. Input is calculated with the direct method (a), an LSF with 100 points uniformly distributed in θ (b), an SVD including gradients of ψ and 7 points (c), an LSF with 100 equidistant points on the desired boundary (d). All equilibria were calculated with FLOW. The points where conditions are imposed are marked in purple.

and $\nabla\psi \cdot \hat{\tau} = 0$ conditions are imposed, Fig. 1(c); the same method and number of points as Fig. 1(b), but using points determined with the “equal distance” approach, more uniformly distributed on the plasma shape, Fig. 1(d). Note that the “tangent” method would give a better approximation with a higher number of N_{fit} points, but no other method could give an “acceptable” result (in the sense of producing a closed $\psi = 0$ curve resembling the desired one) with a comparatively small number of points. Also, using such a low number of points reduces the computational time by a factor of ~ 10 with respect to the one for Fig. 1(b), from ~ 2 seconds to ~ 0.2 seconds (a slow machine was used to test computational speeds; faster runtimes are obtained with a standard desktop or laptop computer). The fixed-boundary equilibrium used for this calculation has a plasma current of 779kA, $\beta_t \simeq 0.11$, $\beta_p \simeq 1.8$, $\ell_i \simeq 0.4$ and $q_* \simeq 2.6$ (see Ref. [20] for the standard definitions of the plasma figures of merit). Free-boundary equilibria have very similar (relative differences $\lesssim 2\%$) properties.

The same methods are tested for a “standard-tokamak” DIII-D equilibrium and the results, qualitatively similar to the ones in Fig. 1, are shown in Fig. 2. The only difference in the methods used for the DIII-D calculations is that 14 points were needed in Fig. 2(c), but for a total of 16 coil currents plus the central solenoid current. For

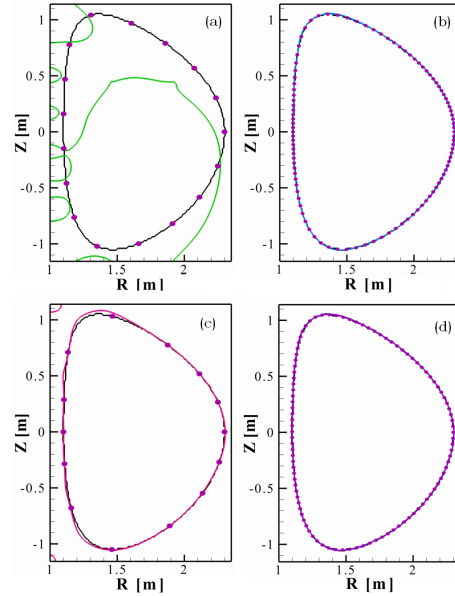


Figure 2. Desired plasma shape (black) and plasma shapes calculated with FREE-FIX input for a DIII-D equilibrium. Input is calculated with the direct method (a), an SVD with 100 points uniformly distributed in θ (b), an LSF including gradients of ψ and 14 points (c), an SVD with 100 equidistant points on the desired boundary (d). All equilibria were calculated with FLOW. The points where conditions are imposed are marked in purple.

this DIII-D equilibrium, the plasma current is 691kA, $\beta_t \simeq 1.1 \times 10^{-2}$, $\beta_p \simeq 0.43$, $\ell_i \simeq 1.4$ and $q_* \simeq 3.25$.

Quantitative error estimates show that “good results” (as identified in Figs. 1 and 2) have average distances from the desired shape ($\langle L_2 \rangle$ norm) of the order of 10^{-4} [m] (typical radii are ~ 1 meter, so that values normalized to the local minor radius are similar to unnormalized ones) and maximum distances ($\langle L_\infty \rangle$ norm) $\lesssim 10^{-2}$ [m] for DIII-D and slightly larger for NSTX. It is found that a point distribution evenly spaced in θ does not perform as well as other strategies. One should be careful in interpreting the numerical results in Figs. 1 and 2, since the accuracy of the free-boundary equilibrium also depends on the equilibrium solver, in addition to the input.

More details on the error estimate for the DIII-D calculation are given in Table 1. The three sections of the table correspond to the methods of Fig. 2) (b), (c) and (d) (in that order) with varying numbers of points. The first line of each section corresponds to the minimum number of points needed to obtain a closed $\psi = 0$ curve in the free-boundary equilibrium calculation. Note that convergence depends on the distribution of the points and on the method (for the “tangent” method, better convergence is obtained with a different point distribution strategy, not shown).

FREE-FIX, matched with the free-boundary version of FLOW, was also tested for several other equilibria, including an NSTX-like elliptical equilibrium, a symmetric, mildly shaped NSTX equilibrium, a DIII-D-like elliptical equilibrium and a DIII-D equilibrium with flow (both toroidal and poloidal), with results qualitatively similar to the ones presented above. It was also verified that if a shape is too different from the design ones for a given set of coils, it may not be possible to find coil currents that will

Method	N_{fit}	point placement	$\langle L_2 \rangle$ [m]	L_∞ [m]
standard	30	equal θ	3.04×10^{-4}	1.66×10^{-2}
standard	50	equal θ	1.77×10^{-4}	8.10×10^{-3}
standard	75	equal θ	1.78×10^{-4}	8.17×10^{-3}
standard	100	equal θ	1.78×10^{-4}	8.18×10^{-3}
“tangent”	13	equal θ	2.39×10^{-3}	1.32×10^{-1}
“tangent”	14	equal θ	7.17×10^{-4}	4.32×10^{-2}
“tangent”	15	equal θ	5.08×10^{-4}	2.22×10^{-2}
“tangent”	20	equal θ	7.54×10^{-4}	3.45×10^{-2}
“tangent”	50	equal θ	2.77×10^{-4}	1.29×10^{-2}
standard	25	equal distance	1.48×10^{-4}	6.73×10^{-3}
standard	30	equal distance	1.62×10^{-4}	7.49×10^{-3}
standard	50	equal distance	1.61×10^{-4}	7.56×10^{-3}
standard	75	equal distance	1.61×10^{-4}	7.60×10^{-3}
standard	100	equal distance	1.61×10^{-4}	7.60×10^{-3}

Table 1. DIII-D error table. Four different measures of error in the obtained shape are shown for several methods and number of boundary points.

allow for the desired plasma shape to be reproduced in a free-boundary equilibrium calculation. More details on these calculations are provided in [14].

In addition to FLOW, FREE-FIX has also been matched to the equilibrium solver of the HYM code [18, 19], which includes energetic particles. Indeed, part of the original motivation for this work was to provide HYM with a tool for calculating free-boundary equilibria given a fixed-boundary one.

4. Conclusions

In this paper, the code FREE-FIX was presented. FREE-FIX calculates the coil currents given a desired plasma shape and plasma equilibrium properties (i.e., a fixed-boundary equilibrium). Coil currents are obtained using a Green’s function method. The Green’s function is known from the literature and expressed in terms of elliptic integrals. The required numerical conditions are that the poloidal magnetic flux due to the plasma and coil currents should have an assigned, fixed value on the plasma boundary, typically $\psi = 0$. The standard procedure is to use a least square fit or singular value decomposition to minimize the difference between the desired and resulting shape in a finite number of points.

Numerical tests for the DIII-D tokamak and the NSTX spherical tokamak were performed with the FLOW equilibrium code, showing good agreement between the desired and obtained shape. A variety of shapes, coil configurations and equilibrium models was tested with success. Remarkably, good agreement can be found with a small number of points, even smaller than the number of unknowns. That is made possible by the innovative idea introduced in this work for matching the desired shape. The new condition is to require that $\nabla\psi$ be normal to the desired plasma shape, i.e. that the $\psi = 0$ curve be tangent to it. One advantage of this numerical method is that the condition on the poloidal magnetic flux gradient only depends on the same elliptic integrals as the $\psi = 0$ condition. Since almost all of the computational cost is in calculating elliptic integrals, this results in a computational cost that is only

about half of the cost of imposing the same number of conditions, but only using the $\psi = 0$ requirement. Moreover, the condition on the derivative proves to be “stronger” than the condition on the value of ψ , in the sense that for the same (small) number of equations a satisfactory result may be obtained when two conditions are imposed in each point, while no good solution is found if only one condition per point is imposed. The choice for the location of the points where the conditions are enforced is of critical importance, in particular for small numbers of points. Also, with some point distribution strategies the “tangent” method will not necessarily outperform the standard method for comparable computational cost when the number of points becomes sufficiently large.

FREE-FIX is freely distributed to the community and it is our hope that other scientists will benefit from its use. An impact may also be made for applications different from the ones described in this paper. In particular, it is suggested that the faster algorithm introduced in this work for determining the input needed for a free-boundary equilibrium could benefit real-time equilibrium reconstructions used for plasma shape and position control and optimization calculations in the design of experiments.

5. Acknowledgments

This work was performed under DOE grant DE-FG02-93ER54215 and PPPL subcontract S013799-F. Our thanks go to E. Belova for bringing the problem to our attention, to S. Gerhardt for his help with NSTX data, to S. Jardin, A. Cerfon, J. Hanson and J.P. Freidberg for useful discussion and to C. Sovinec for his encouragement.

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